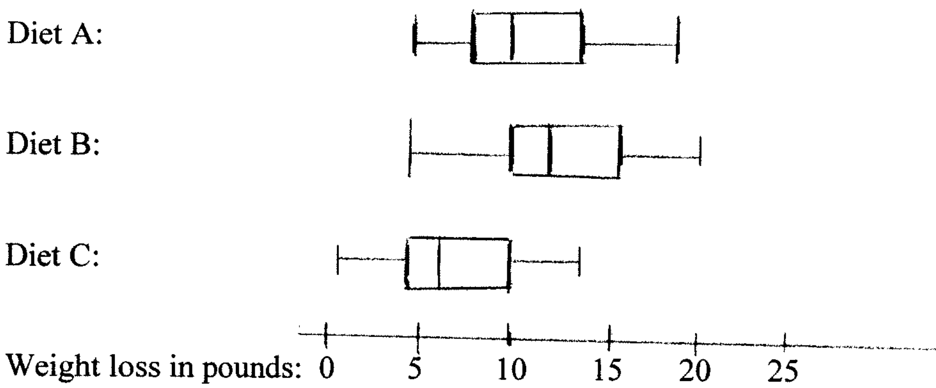
3/8/07
Part II

Masters Exam: APPLIED PART Answer all 3 questions. Show work in bluebook.

(30 Points) 1) A diet specialist wanted to compare three diets: Diet A , Diet B, and Diet C. The specialist took 180 people and randomly assigned 60 people to Diet A, 40 people to Diet B, and 80 people to Diet C. Each person was weighed at the start of the diet, and again after four months on the diet. The response variable was the number of pounds loss under the diet. The following are the box and whisker plots for the three diets:



Carry out a test at the 0.05 level of significance of the null hypothesis that there is no difference between the three diets in terms of the chance that a person will lose at least 10 pounds. The alternative hypothesis is that there is some difference between the three diets in terms of the chance that a person will lose at least 10 pounds.

(30 Points) 2) A researcher was interested in studying the association between the height of an individual and the size of the individual's eye pupil. A sample of 25 people were picked and each person's height was measured in inches, and the diameter of their eye pupil was measured in millimeters. For the group of 25 people, the following 9 statistics were computed:
Mean height = 66 inches, Median height = 63 inches, standard deviation of height = 3.3 inches; coefficient of relative variation = 0.05
Mean diameter of pupil = 8.8 millimeters (mm), Median diameter of pupil = 8.7 mm, standard deviation of diameter of pupil = 1.1 mm, coefficient of relative variation = 0.125
The Pearson product-moment correlation between height and diameter of pupil = 0.80.

a) Give the least-squares line for predicting an individual's pupil diameter in mm from their height in inches.
b) You have available the 9 summary statistics, but the original individual observations are not available. You want height in millimeters instead of inches. (One inch = 25.4 mm). You also want the size of the individual's eye pupil to be measured by its area in square millimeters, where area = $\pi\,(0.5 \text{ Diameter})^2$. For each of the nine summary statistics given above, give where possible the new values under the change of scales of measurement. If for a particular statistic this is not possible, then say so.

(40 points) 3) An educational researcher seeks to study the effect of training teachers to use video educational materials, on the performance of the teachers in the classroom. The researcher wants to see if the training has an effect, and to test whether the effect is the same for elementary as for Junior High school teachers. Four elementary schools, and four Junior High Schools have agreed to participate in the study. The specific schools used in the study were not picked at random from a population of schools, but were rather fixed for feasibility and administrative control by the experimenter.

Within each of the 8 schools, there will six groups of teachers. Even though each group has several teachers, the experimental units are Groups. Groups is regarded as a random effect. The 6 groups within a school will be randomly assigned, 3 groups to video training, and 3 groups to the control. . For each Group the researcher will compute the response variate y, as the mean group performance on a normally distributed scale.

The experimental design involves the following factors:

	Factor	Levels of Factor
Factor A	Type of School	A_1: Elementary; A_2: Junior High
Factor S	School	S_1 S_2 S_3 S_4 S_5 S_6 S_7 S_8
Factor T	Treatment type	T_1: Video training ; T_2: Control
Factor G	Group	G_1 to G_{48}

The data layout for design is the following:

	A_1 Elementary				A_2 Junior High			
	S1	S2	S3	S4	S5	S6	S7	S8
T_1 Treatment	G_1	G_7	G_{13}	G_{19}	G_{25}	G_{31}	G_{37}	G_{43}
	G_2	G_8	G_{14}	G_{20}	G_{26}	G_{32}	G_{38}	G_{44}
	G_3	G_9	G_{15}	G_{21}	G_{27}	G_{33}	G_{39}	G_{45}
T_2 Control	G_4	G_{10}	G_{16}	G_{22}	G_{28}	G_{34}	G_{40}	G_{46}
	G_5	G_{11}	G_{17}	G_{23}	G_{29}	G_{35}	G_{41}	G_{47}
	G_6	G_{12}	G_{18}	G_{24}	G_{30}	G_{36}	G_{42}	G_{48}

In ANOVA table below, for each Sources of variability (SV) write out degrees of freedom (df), indicate whether SV is a fixed or random effect, give Expected Mean Square, and write F-ratios in terms of Mean Squares (e.g. for term C, F-ratio = MS_C/MS_D).

SV	df	F or R	EMS	F-ratio
A				
S|A				
T				
T x A				
TS|A				
G|TS|A				